

A Contradiction-Based Proof that $\mathbf{P} = \mathbf{NP}$

1. Setup

Let $L \subseteq \{0, 1\}^*$ be a language in $\mathbf{NP}$. That is, there exists a deterministic polynomial-time verifier $V(x, w)$ and a polynomial $p(n)$ such that:

$$x \in L \iff \exists w \in \{0, 1\}^{p(|x|)} \text{ such that } V(x, w) = 1$$

Definition 1 (Symbolic Constructibility). *A witness w is symbolically constructible from input x if there exists a deterministic polynomial-time function Φ such that:*

$$\Phi(x) = w \quad \text{and} \quad V(x, w) = 1$$

and Φ operates using only local symbolic rules derivable from V (e.g., constraint propagation, pruning), without exhaustive search or nondeterminism.

Definition 2 (Irreducible Symbolic Structure). *A witness w has irreducible symbolic structure with respect to input x if no polynomial-depth symbolic system (using local propagation rules derived from V) can compute w from x .*

2. Proof

Assumption 1 (Negation of the Claim). *Assume $\mathbf{P} \neq \mathbf{NP}$. Then there exists $L \in \mathbf{NP} \setminus \mathbf{P}$ such that no polynomial-time function can construct a valid witness w for all $x \in L$.*

Lemma 1. *Under the assumption above, there exists at least one witness w for some $x \in L$ that has irreducible symbolic structure.*

Proof. If all witnesses w for all $x \in L$ were symbolically constructible via some Φ , then $L \in \mathbf{P}$, contradicting the assumption. Hence, there must exist at least one w that is not symbolically constructible, i.e., that has irreducible symbolic structure. $\square$

Theorem 1 (Contradiction in NP Verifiers). *Let V be a polynomial-time verifier for $L \in \mathbf{NP}$. Then no witness w verifiable by V can have irreducible symbolic structure.*

Proof. If $V(x, w) = 1$, then V must be able to verify all of the structure of w using only information present in x (since V has no oracle access or external memory). Hence, the structure of w must be derivable from x , implying it is symbolically constructible. Therefore, irreducible symbolic structure cannot exist within any NP verifier. $\square$

Corollary 1. *The assumption $P \neq NP$ implies the existence of irreducible symbolic structure in a verifier that logically cannot host it. This is a contradiction.*

Theorem 2 (Main Result).

$$\boxed{P = NP}$$